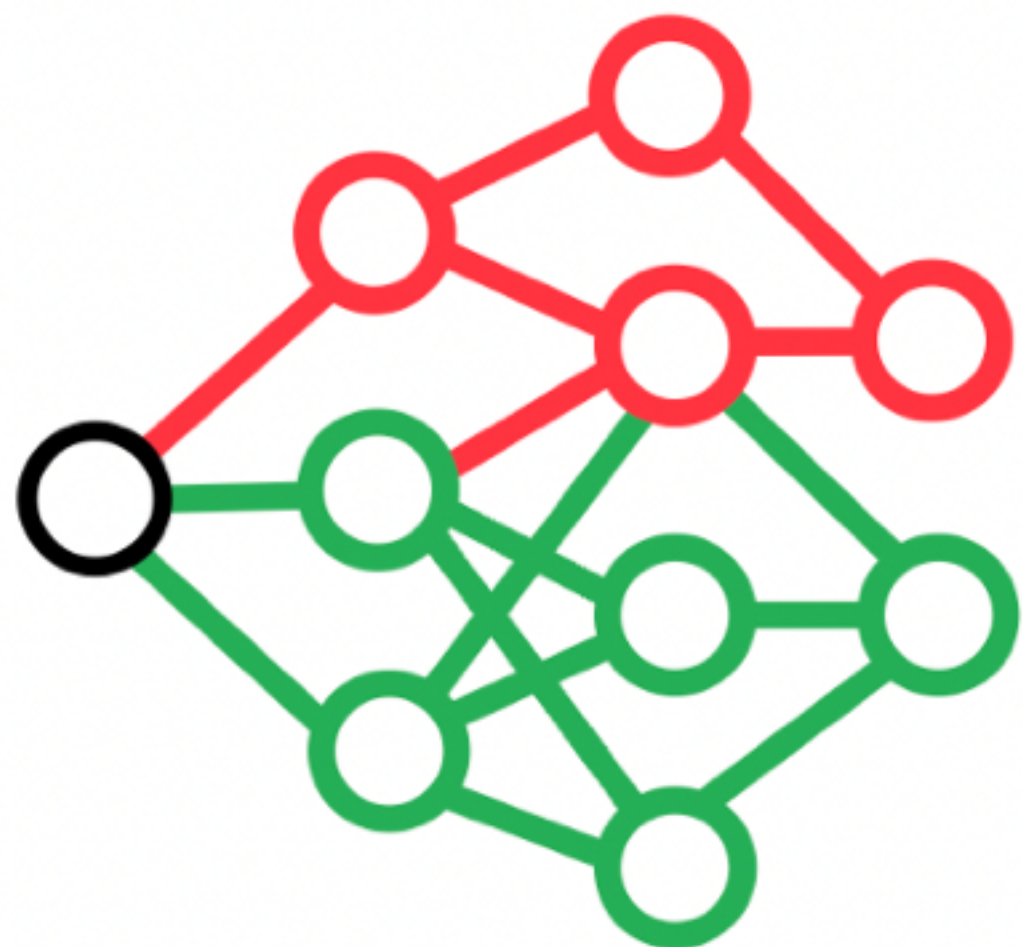


○ Harmless States

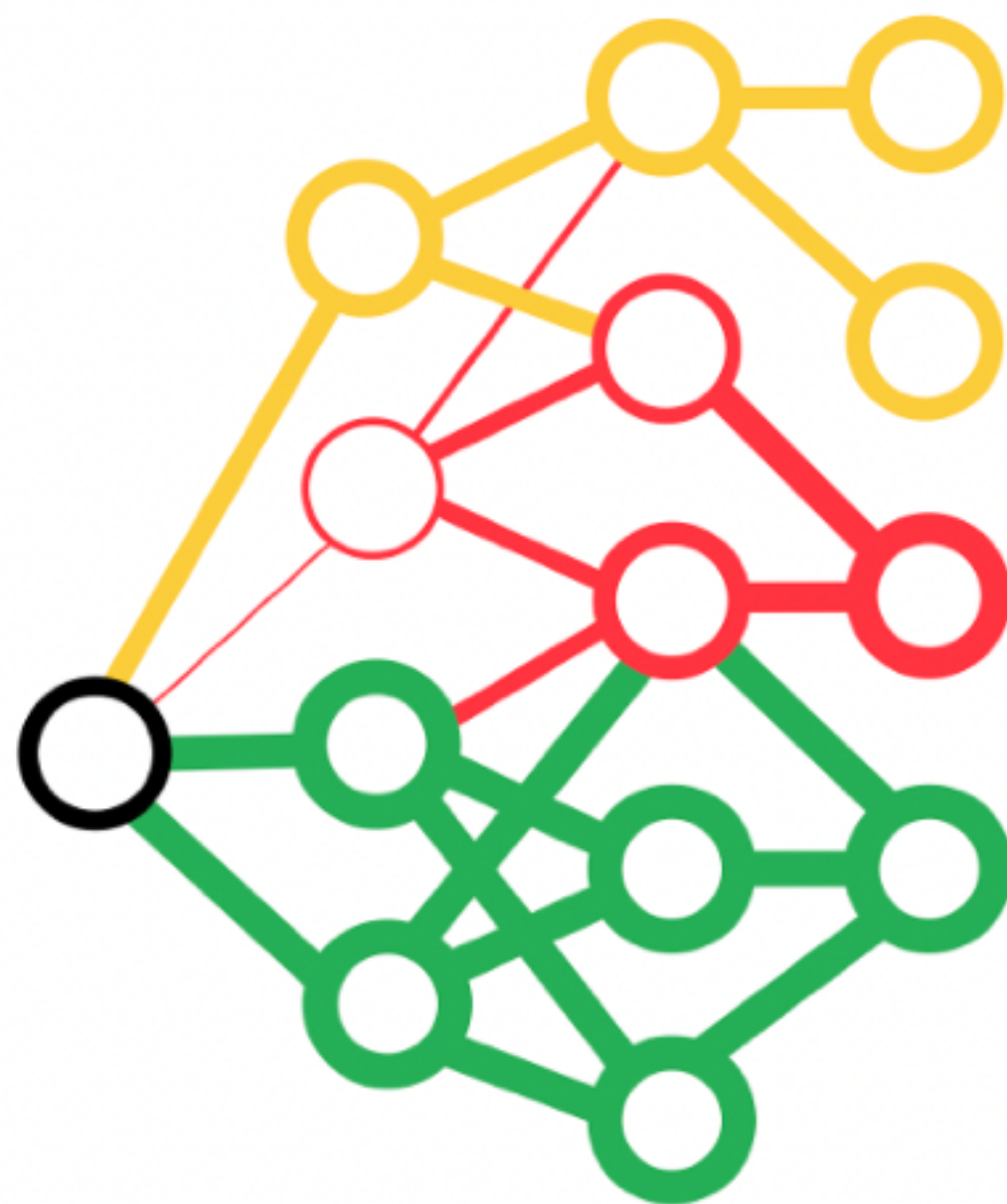
○ Harmful States

○ Refusal States

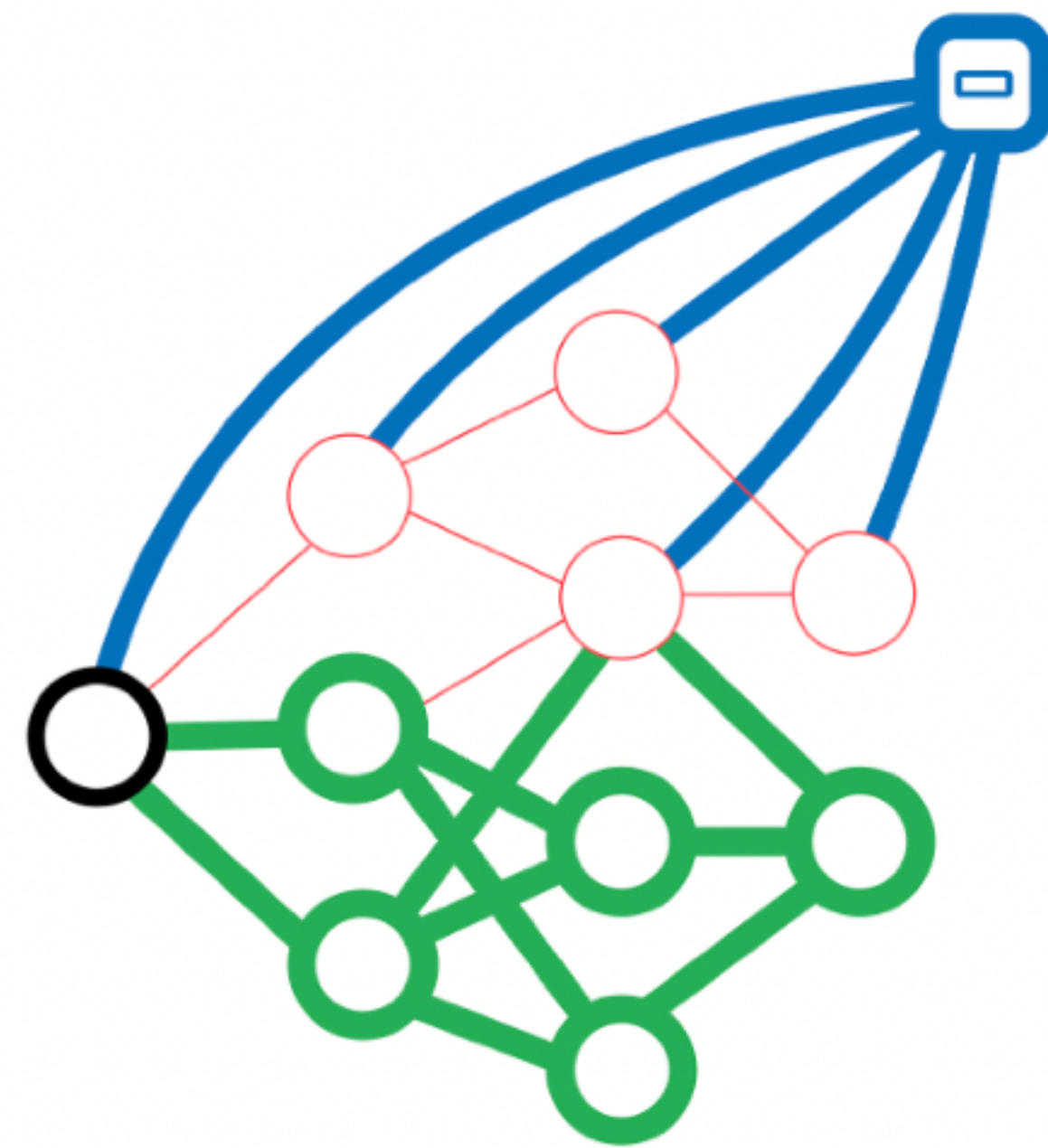
☐ Circuit Breakers



Instruct Model



Refusal Training



Circuit Breaking

$$\mathcal{L}_s = \text{ReLU}(\text{cosine_sim}(\text{rep}_{\mathcal{M}}(x_s), \text{rep}_{\mathcal{M}_{\text{cb}}}(x_s))) \quad \triangleright \text{RR Loss}$$

$$\mathcal{L}_r = \|\text{rep}_{\mathcal{M}}(x_r) - \text{rep}_{\mathcal{M}_{\text{cb}}}(x_r)\|_2 \quad \triangleright \text{Retain Loss}$$

$$\mathcal{L} = c_s \mathcal{L}_s + c_r \mathcal{L}_r \quad \triangleright \text{Loss to be Optimized}$$

Source: (Circuit breaking; Zou et al., 2024).

Circuit breaking

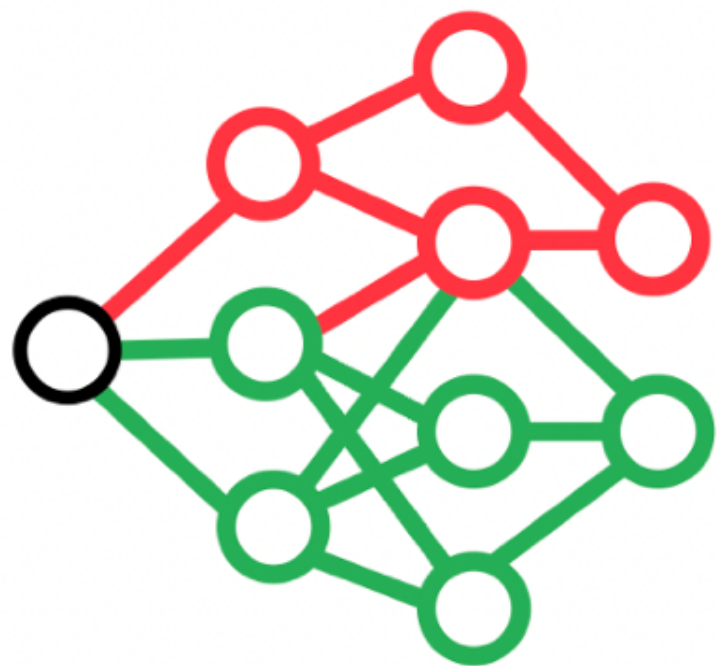
Jailbreaking defenses

 Harmless States

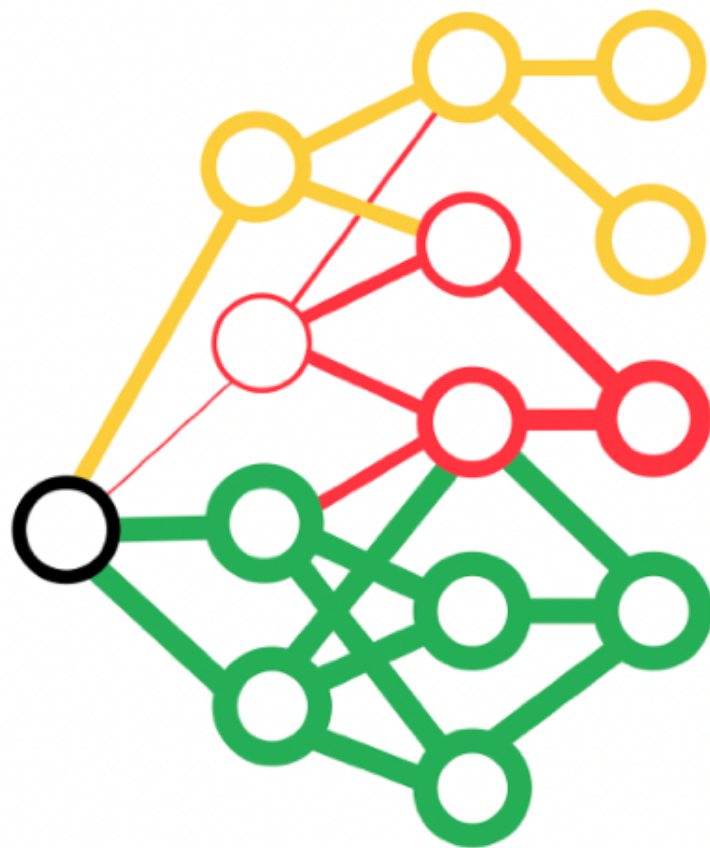
 Harmful States

 Refusal States

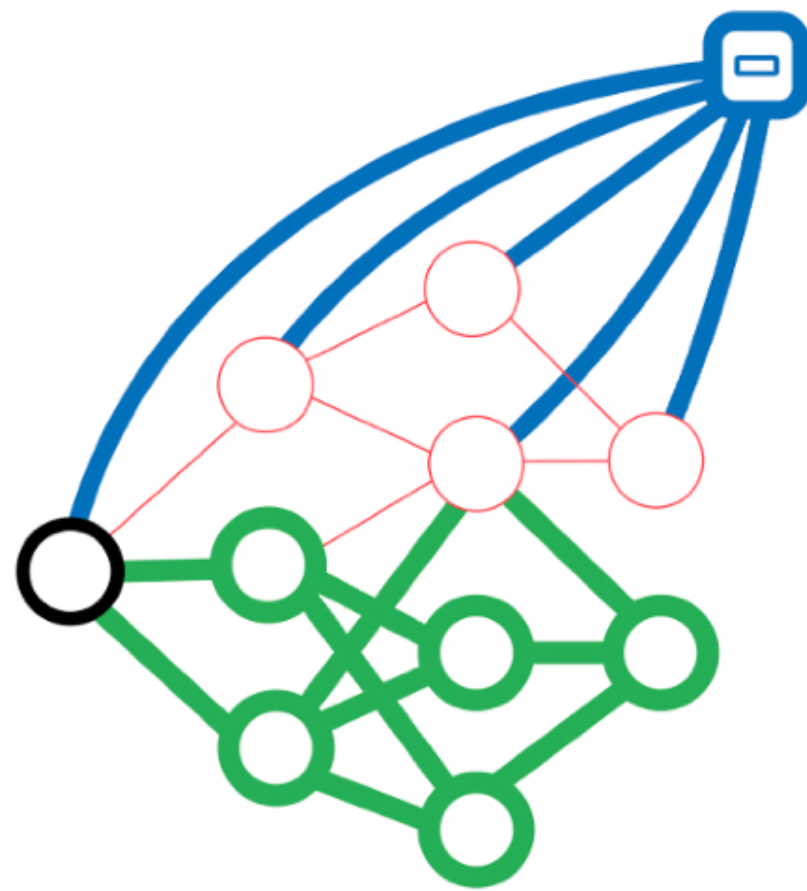
 Circuit Breakers



Instruct Model



Refusal Training



Circuit Breaking

Jailbreaking defenses

Circuit breaking

○ Harmless States ○ Harmful States ○ Refusal States  Circuit Breakers

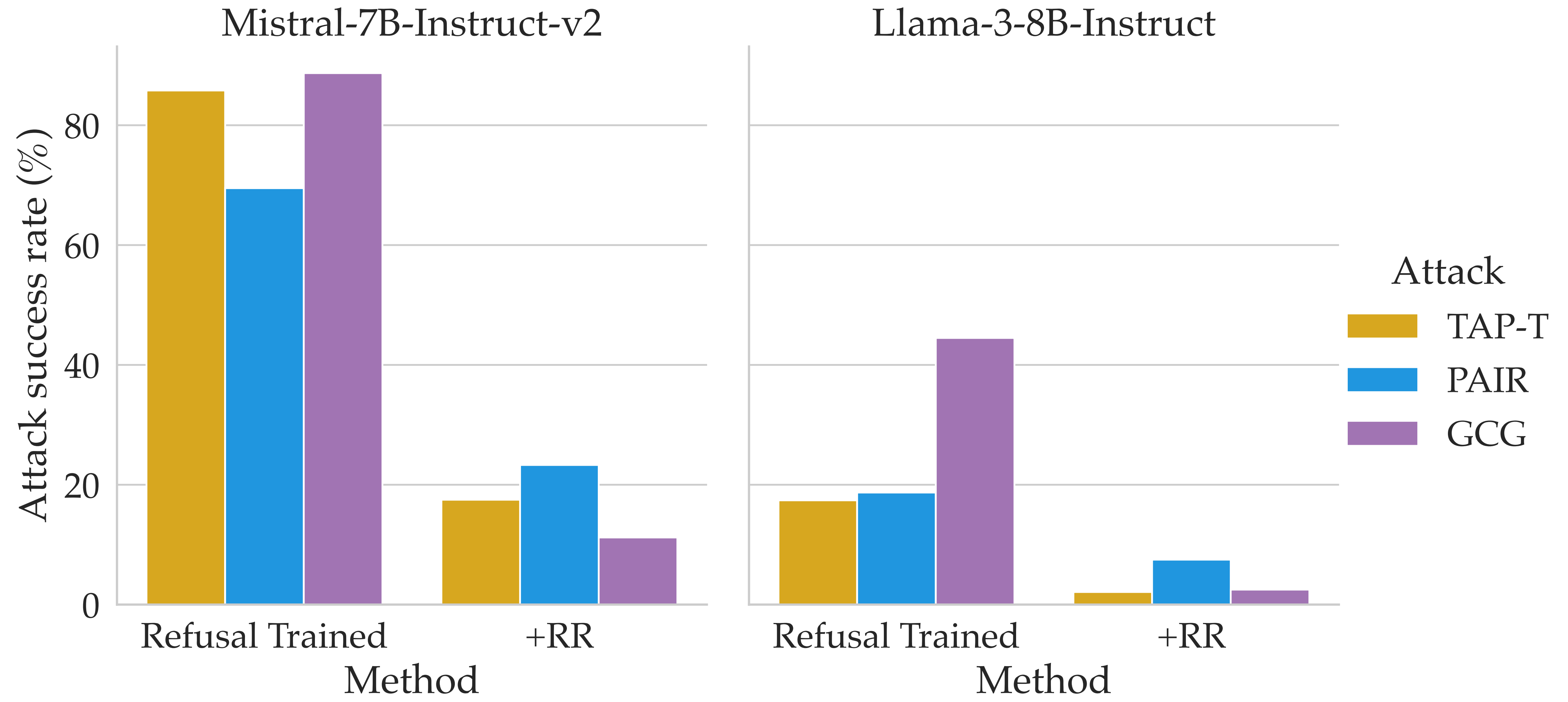


$$\mathcal{L}_s = \text{ReLU}(\text{cosine_sim}(\text{rep}_{\mathcal{M}}(x_s), \text{rep}_{\mathcal{M}_{\text{cb}}}(x_s)))$$
$$\mathcal{L}_r = \|\text{rep}_{\mathcal{M}}(x_r) - \text{rep}_{\mathcal{M}_{\text{cb}}}(x_r)\|_2$$
$$\mathcal{L} = c_s \mathcal{L}_s + c_r \mathcal{L}_r$$

- ▷ RR Loss
- ▷ Retain Loss
- ▷ Loss to be Optimized

Jailbreaking defenses

Circuit breaking



Source: (Circuit breaking; Zou et al., 2024).